

Research Proposal: Detecting Insider Threats through LLM-Based Embeddings and Reinforcement Learning

1. Introduction

Insider threats represent a critical security challenge for modern organizations. Traditional detection systems rely on static machine learning models trained on historical data, but insider threat behaviours evolve over time as adversaries adapt their tactics. This research proposes an adaptive framework that combines Large Language Model (LLM) embeddings with reinforcement learning to dynamically identify suspicious email communications.

The key idea is treating threat detection as an online learning problem. Rather than deploying a fixed classifier, we develop an intelligent agent that continuously learns from feedback, improves its detection policy, and actively seeks clarification on uncertain cases.

2. Research Objectives

This research addresses three interconnected objectives:

Objective 1: Convert Raw Emails to Meaningful Representations

Our first task is to extract semantic features from unstructured email text. We will use pre-trained LLMs (such as BERT) to generate high-dimensional vector representations (embeddings) that capture key characteristics like sentiment, urgency, and deviations from a user's normal communication patterns.

Objective 2: Build an Adaptive Detection Agent

We will implement a Contextual Multi-Armed Bandit algorithm (specifically LinUCB) that learns to classify emails as threats or benign. Unlike static models, this agent balances two competing needs: exploiting known patterns versus exploring uncertain cases to reduce future errors.

Objective 3: Evaluate Performance on Realistic Data

We will test the system using the CERT Insider Threat Dataset and measure how effectively the agent detects threats while minimizing analyst workload.

3. Methodology

Phase 1: Email Feature Engineering Using LLM Embeddings

Approach:

Process each email through a pre-trained transformer model (BERT or similar)

Extract vector representations from the model's hidden layers

Apply dimensionality reduction to create a compact, efficient feature space

Phase 2: Implementing the Reinforcement Learning Agent

Approach:

Deploy a LinUCB bandit algorithm as the decision-making engine

The agent maintains a linear model relating email features to threat probability

The agent calculates uncertainty estimates for each decision

When uncertainty is high, the agent flags emails for human review to gather training signal

Phase 3: Training and Evaluation

Approach:

Simulate a Security Operations Centre environment using the CERT dataset

Allow the agent to process emails sequentially, receiving feedback labels

Update the agent's policy after each decision using efficient online learning rules

Measure performance using cumulative regret (total cost of incorrect decisions)

4. Evaluation Criteria

The system will be assessed using three primary metrics:

Regret Minimization: Does the agent's detection performance improve over time, approaching an optimal oracle?

Detection Accuracy: What is the true positive rate versus false positive rate at different operating points?

Resource Efficiency: Can the agent identify threats while respecting analyst capacity constraints?

5. Data and Resources

The CERT Insider Threat Dataset (v6.2) will serve as our primary data source, providing realistic organizational communication logs with labelled malicious events.